

Visualizing and Predicting trend in global Life Expectancy from 2001 to 2019

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Abstract—What factors drive longevity. Why are some island nations living longer than some developed nations. Longevity and life expectancy is a core issue that touches on the fundamental of livelihood, it drives immigration and emigration for a search of better quality of life, longevity contributes to brain drain of a region in search for a better opportunity to maximize the quality of a individual or family life. This project aims to explore this issue thoroughly by analyzing the data from world bank via kaggle. The goal is to visualize which factors influences or drive longevity, predict the longevity of each dataset using machine learning and compare it to the actual value reported and suggest recommendations for future work based on the data interpolation.

Index Terms—Life Expectancy, Public Health, Quality of Life

I. INTRODUCTION

Our project will analyze the trend and increase in life expectancy across the world during the pre-Covid era ranging from 2001- 2019. Factors to be examined are socioeconomic factors such as country investments in healthcare per GDP, education spending per GDP, environmental management, region and various health metrics such as sanitation and diseases

The overall goal of the project is to explore public health and the role of divergent factors on life expectancy. The intended audience includes everyone from policy makers who make policy on budget allocation, students (the role of education in life expectancy) and the public who vote for legislators that enacts and oversees this body.

Our hope is to guide our intended audience on the role and interconnection of each factors and highlight the most important factor or metrics that should be prioritize for improvement per region to maximize life expectancy.

II. DATASET AND FEATURES

A. Data Description

The structured dataset consists of categorical and numerical variable. The columns are stratified into: "Country Name", "Country Code", "Region", "Income Group", "Year", "Life Expectancy World Bank", "Prevalence of Undernourishment", "CO2", "Health Expenditure" %, "Education Expenditure", "Unemployment", "Corruption", "Sanitation", "Injuries", "Communicable" and "Non-communicable".

B. Data Information

The name of the dataset is life expectancy and socio-economic (world bank)

- Source: Kaggle
- Link: https://www.kaggle.com/datasets/mjshri23/life-expectancy-and-socio-economic-world-bank?utm_source=chatgpt.com

C. Data Size

- No of rows are 3307
- No of columns are 16
- range: 2001 – 2019
- Country Name: Unique variable is 174
- Country Code: Unique variable is 174
- Region: Unique variable is 7
- Income Group: Unique variable is 7
- Year: Missing 0, Mismatched 0
- Life Expectancy World Bank: Missing 188 variable
- Prevalence of Undernourishment: Missing 684
- CO2: Missing 152
- Health Expenditure %: Missing 180
- Education Expenditure: Missing 1090
- Unemployment: Missing 304
- Corruption: Missing 2331
- Sanitation: Missing 1247

D. Data Initial Observation

Before performing any imputation to fill missing values. Some columns such as (corruption) were found to consist of a high amount of missing values and were excluded from analysis. A quick overview of the dataset saw an increase in life expectancy as the year progress. Also observed were the exclusion of some countries in the dataset which will slightly alter the global results.

E. Data Imputation

Missing values were filled based on the mean or median per region reflecting geographic similarities among nearest neighboring countries. The column corruption was dropped due to a significant high amount of missing values.

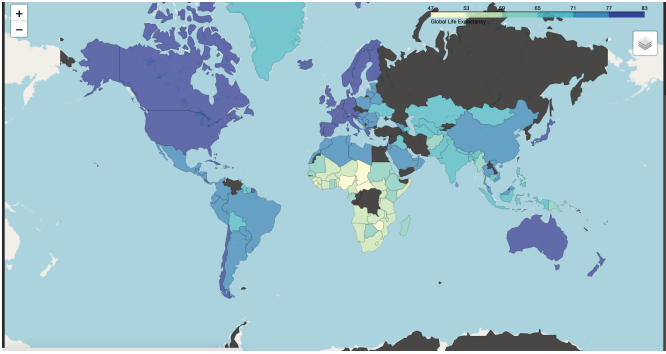


Fig. 1. World Map color showing the life expectancy across all provided region after imputation

Missing datum from countries missing specific datum in a row in a particular year were filled using forward fill and backward fill.

III. METHODS

A. ML Algorithms : Predictive modeling and DEEP Learning KERAS

- Predictive Modeling: Linear Regression, KNeighbor Regressor, Decision Tree Regressor, Random Forest Regressor and Gradient Boost Regressor, XGBoost.
- Deep learning KERAS Parameters: Adam Optimizer, Exponential and Polynomial Decay, Early stopping, ReduceLROnPlateau, Sequential.
- Train test split of 80: 20, Train test split histogram to determine best train test split proportion.
- Evaluation Metrics: MAE and R square.
- Feature Metrics: SHAP on Linear Regression.

B. Sci-Kit Learn Algorithms and Hyper-parameters

- xgboost (learning_rate= 0.1, n_estimator= 100, min_child_weight = 5, max_depth= 2),
- KNeighbor Regressor (n_neighbors= 1, 10, 100, weights= uniform),
- Decision Tree Regressor (max_depth= 2, 5, 10),
- Random Forest Regressor (n_estimator= 10, 100, max_depth= 5, 10),
- Gradient Boosting Regressor(n_estimator= 100, learning_rate=0.1, max_depth = 2),
- xgb.XGBRegressor(max_depth= 2, learning_rate= 0.1, n_estimators= 100, min_child_weight= 5)

C. Cross Fold Validation Hyper-parameters

- Cross Validation and Cross_Fold = (kFold = n_splits: 5, shuffle = True, scoring = negative_mean_absolute_error)

D. GRIDSearch CV ML Hyper-parameters and Explanation

- KNeighborsRegressor": (n_neighbors: [1, 5, 10, 20, 50], weights: [uniform, distance])
- RandomForestRegressor(n_estimators: [10, 50, 100], max_depth: [3, 5, 10], min_samples_split: [2, 5, 10], min_samples_leaf: [1, 2, 5, 10]),

- GradientBoostingRegressor(n_estimators: [50, 100, 200], learning_rate: [0.01, 0.1, 0.2], max_depth: [2, 3, 5], min_samples_leaf: [1, 2, 5, 10]),
- DecisionTreeRegressor(max_depth: [2, 3, 5, 10], min_samples_split: [2, 5, 10], min_samples_leaf : [1, 2, 5, 10]),
- LinearRegression(fit_intercept: [True, False]),
- xgb.XGBRegressor(n_estimators: [50, 100], learning_rate: [0.01, 0.1, 0.2], max_depth: [2, 3, 5], min_child_weight: [1, 2, 5])

E. KERAS Hyperparameter

- Epoch = 500,
- Batch size = 256,
- Decay fraction = 0.8,
- Initial learning rate = 0.05, End learning rate = 0.001, Power = 2.0, Decay_steps = decay_epochs * steps_per_epoch
- Early stopping Monitor = val_loss, Patience = 100, Restore_best_weights = True
- Optimizer = Adam optimizer,
- Loss = Mean Absolute Error (MAE),
- Dropout rate = 0.1,
- Reduce learning rate Monitor = validation loss, Factor = 0.5, Batch_size = batch_size, Patience = 30, Min learning rate = 0.000006, Verbose = 1
- Activation = RELU

F. Parameter Explained

Sci-Kit Learn Regression Parameters:

- max depth: vertical complexity of the tree (nodes)
- n estimator: size of the forest or tree (number of trees)
- learning rate: scaling factor of each new tree
- child weight: regularization used to prevent overfit-
- samples split: regularization that control node growth before splitting to child nodes
- min samples leaf: minimum sample of leaf node required to have, smoothness vs spiky
- weights: used in Kneighbor to determine influence of Kneighbor on final prediction
- Mean Absolute Error: measure the magnitude of error using actual vs predicted value
- R²: measuring variance in a variable

KERAS Parameters

- Adam Optimizer: train neural network, analyze gradients to understand momentum and scale.
- Initial Learning rate: starting step for the optimizer to use.
- Early Stopping: regularization used to prevent overfitting.
 - Patience: grace period for model to improve before termination
- Dropout Rate: regularization used to prevent over reliant on a neuron layer
- Epoch: Amount of time the model have studied the dataset

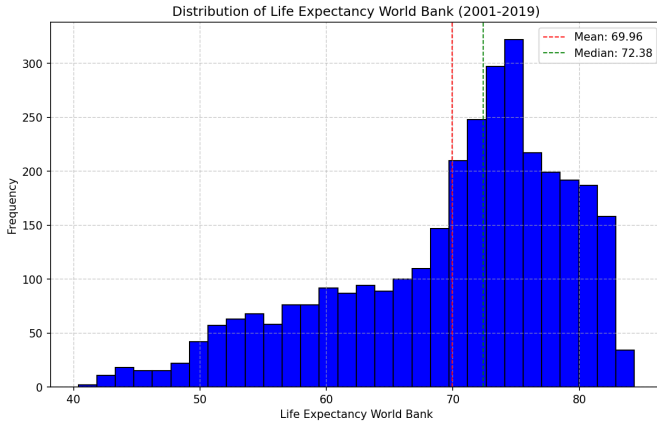


Fig. 2. Histogram plot of average global median of life expectancy

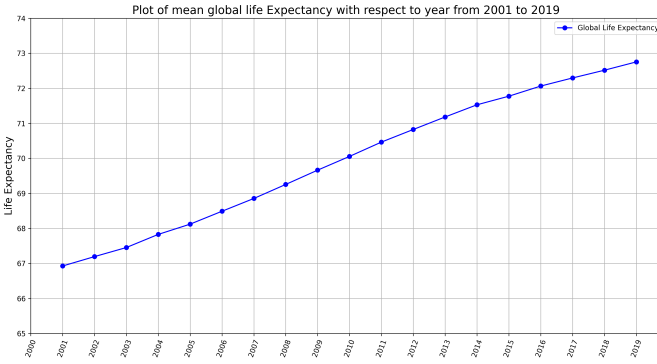


Fig. 3. Line graph shows an increase trend in Global life Expectancy

- Activation: determine the activation of neuron (fire or not) for learning complex patterns like waves, curves, etc.
- Decay fraction: combination of learning rate decay and learning rate schedule
- Learning rate decay: used to reduce the size of the updates a model weights make while training
- Learning rate schedule: adjusting the learning rate during the training process
- Power: polynomial decay, for reaching a specific end goal over a number of steps which determine the drop of (linear = 1)
- Batch size: training samples processed either forward or backward
- Loss function : measure how wrong the model prediction are compared to actual target (MAE, MSE)

IV. EXPERIMENTS AND RESULTS

A. EDA Visualization Result

- Distribution of life expectancy from 2001 to 2019 showed the mean and median age globally was 69.96 and 72.38, this is excluding crucial data from countries such as Korea with a life expectancy of over 84 as of 2026. This was shy of the life expectancy of 73.1 provided by the WHO which include the complete data of all countries.

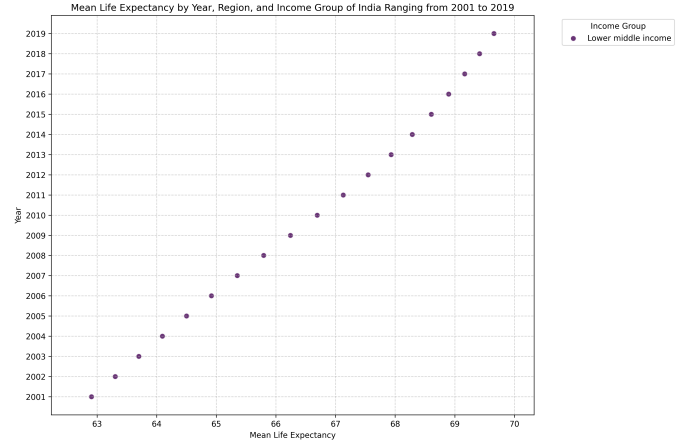


Fig. 4. Mean Life Expectancy from 2001 to 2019 using data derived from India

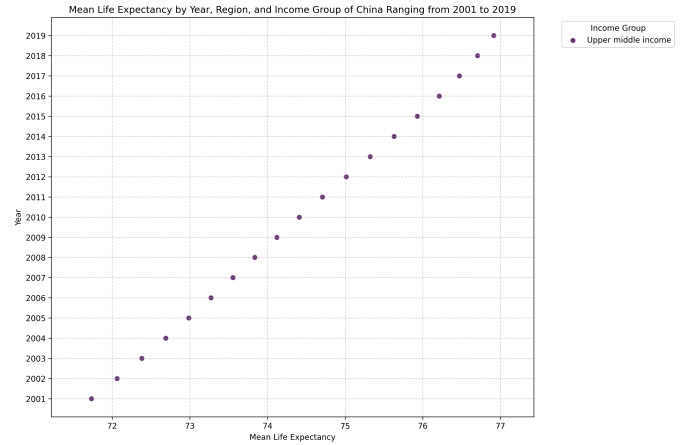


Fig. 5. Mean Life Expectancy from 2001 to 2019 using data derived from China

- Using a random country (India in a specific region, country and region can be changed in DEMO), the life expectancy of India was found to have increased from 62.9 to 69.56 between that time span an increase of 7 points or 7 years.
- Similar trend was observed in China as seen in Figure 4.
- Surprisingly sanitation had a weak influence on life expectancy as seen by the graph among Latin American and Caribbean countries. No visible trend was observed which indicates a mastery or proficient handling of management, separation and disposal of human excreta to prevent disease transmission, this is obvious as disease like cholera and typhoid that were killers in the early 20th century and before are no longer a major factor in life expectancy.
- A similar trend was found in countries located in South East Asia
- Comparing a region such as sub-Saharan Africa (3 countries were examined based on geographic proxim-

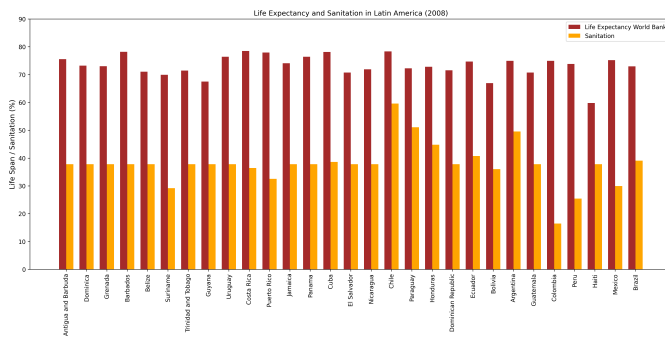


Fig. 6. Life expectancy using sanitation as a factor in Latin America and the Caribbean

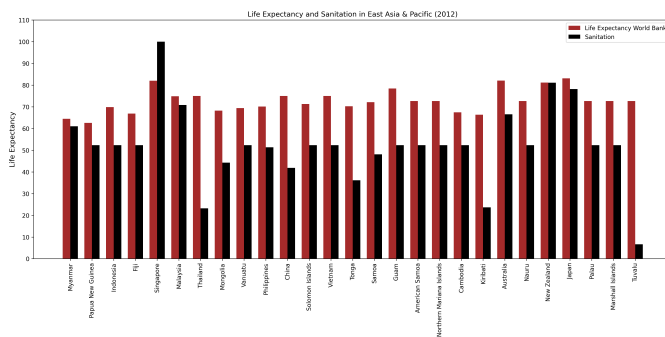


Fig. 7. Life expectancy using sanitation as a factor in South East Asia for the year 2012

ity (neighbors). Ethiopia, Eritrea and Sudan shared a similar border. The life expectancy of all 3 countries were compared from 2001 to 2019. Surprisingly despite huge regional tension, there was an upward trend in life expectancy seen in all 3 countries mirroring the WHO analysis of a global increase in trend. Sudan rose a total of 7 years (58 to 65.3), Ethiopia rose a total of 13 years (52 to 66.5), Eritrea rose a total of 11 years (55.8 to 66.3) despite all 3 countries characterized as low income based on their GNI per capita as seen in Figure 8.

- Comparing prevalence of undernourishment (malnutrition) as a metric to life expectancy using China as an example. A negative correlation between undernourishment and life expectancy as undernourishment decreased, life expectancy generally increased from (71.32 to 76.92) as seen in Figure 9.
- A similar trend was observed in Singapore who boast one of the highest life expectancy globally as seen in Figure 10.
- Comparing undernourishment in South Asia, countries with the highest prevalence of undernourishment often had a lower life expectancy despite seeing an upward increase as seen in Figure 11
- Similarly, life expectancy increased when comparing prevalence of undernourishment in middle east and North Africa, Israel had a lower prevalence of undernourishment and had one of the highest life expectancy in that region

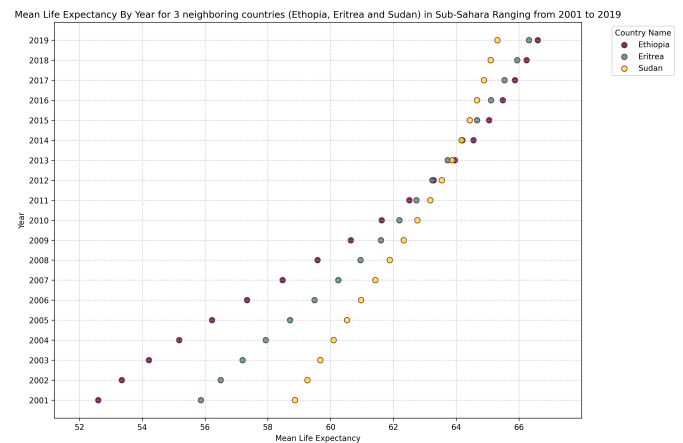


Fig. 8. Comparing the mean life expectancy for 3 neighboring countries in Sub-Sahara Africa

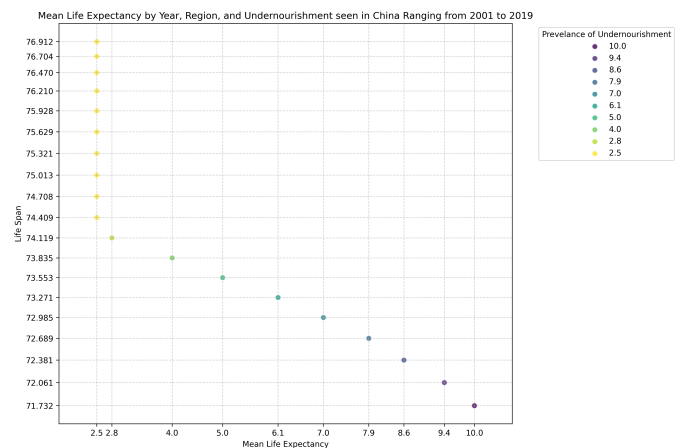


Fig. 9. Trend in life expectancy as prevalence of undernourishment decreases in China from 2001 to 2019

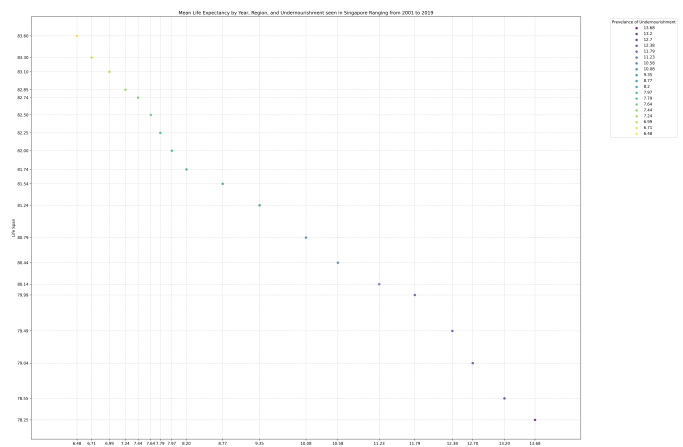


Fig. 10. Trend in life expectancy as prevalence of undernourishment decreases in Singapore from 2001 to 2019

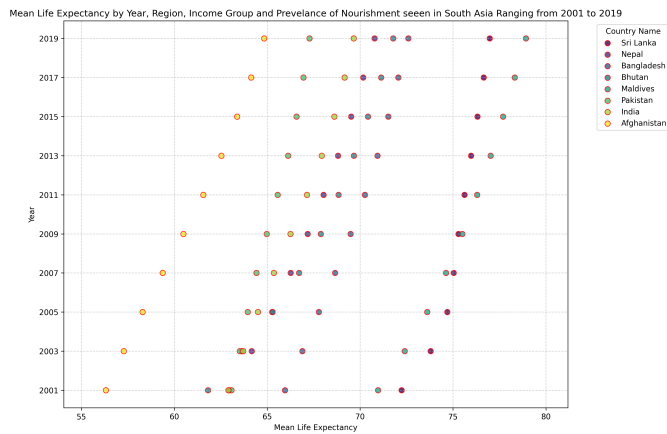


Fig. 11. Comparing the mean life expectancy trend for all countries located in South Asia from 2001 to 2019

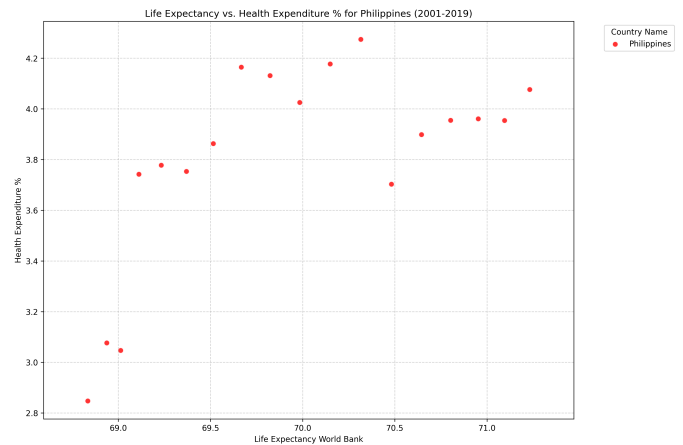


Fig. 13. Scatter plot showing the trend in life expectancy in Philippines based on allocated health expenditures per GDP capita.

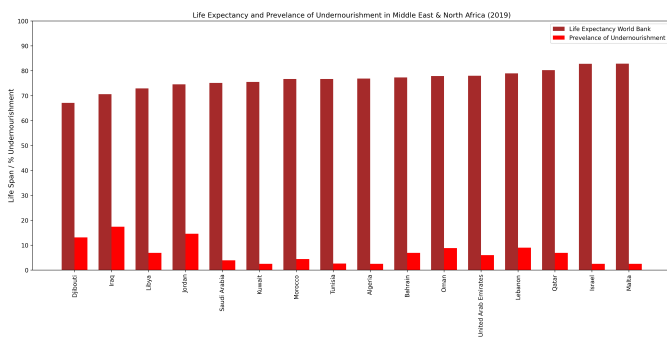


Fig. 12. Bar graph comparing the mean life expectancy trend taking into account prevalence of undernourishment in the Middle East and Northern Africa for the year 2019

as seen in Figure 12.

- Comparing health care expenditure using Philippines, there was a general positive correlation between life expectancy and amount of GDP spent on health care as seen in Figure 13. An exception was noted around 2013/2014 when there was drop, in health care expenditure but a modest increase in Life expectancy, Further analysis shows that the Philippines had one of the costliest and deadliest natural disaster in the history of their country (Typhoon Hagupit) and Super Typhoon Yolanda.
- Factoring communicable disease (STD and disease spread around) and non-communicable disease (cancer) in the region East Asia pacific. A log scale was used to show the amount of GDP expenditure in comparison to the life expectancy as seen in Figure 14. The result show no trend in amount of GDP spent in health care and life expectancy since each country GDP in the east-asia pacific region is based on capital output. A 3% in a low GDP nation does not necessarily correlate to a 2% in a high GDP nation.
- The box plot showing the mean, median and IQR range of all region showed that Sub-Saharan Africa had the lowest mean of 57.5 while North America had the highest

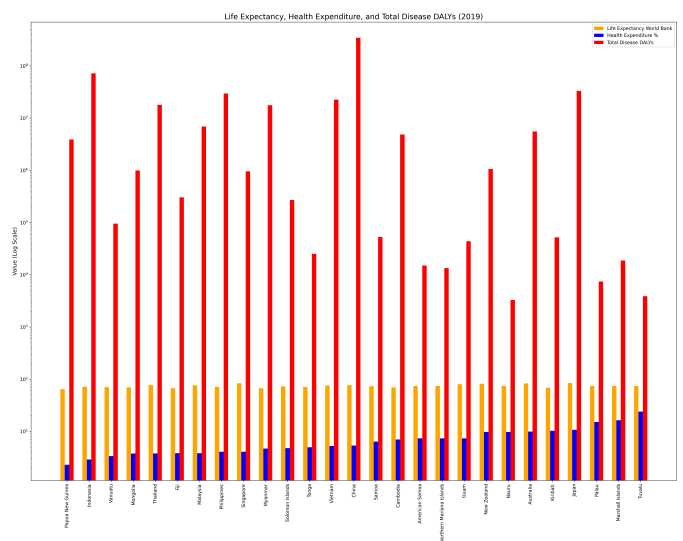


Fig. 14. Log Bar graph comparing the mean life expectancy trend taking into account total disease (communicable and noncommunicable) in the East Asia and Pacific Region

mean of 79.68, beating out countries that have the highest global life expectancy in their region such as Japan or Singapore (indicating a bit of uniformity in the North America Region), Europe and Central Asia was a close second at 76.18, as seen in Figure 15

- Life expectancy typically increases across all region for instance, observing the life expectancy of all region showed that sub-Sahara Africa region had a life expectancy of 61.5. However, the average life expectancy is 57.5 which is a culmination from 2001 to 2019 as seen in Figure 16. It is anticipated that bearing everything in mind and the absent of natural disaster, this trend should increase in the next 20 years but disproportionately unless underlining issues are addressed.
- Finally, life expectancy per year globally from 2001 to 2019 show a box plot depicting a gradual increase across

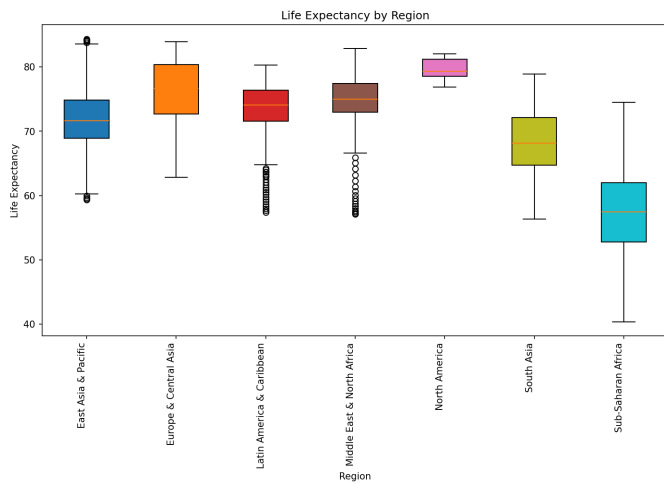


Fig. 15. Box plot showing the average global life expectancy per region from 2001 to 2019, the whiskers represent the maximum and minimum life expectancy for a specific region during that duration, the yellow line indicates the median global life expectancy.

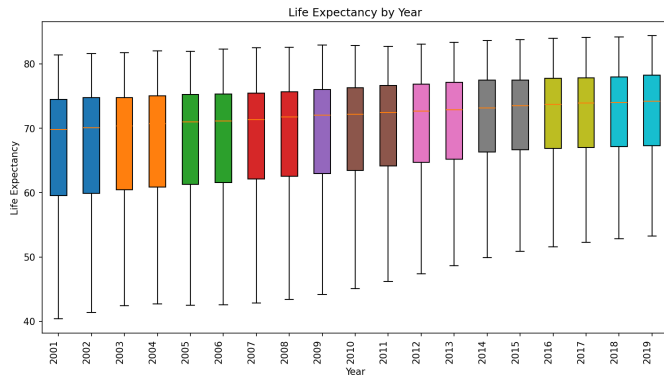


Fig. 16. Box plot showing the average global life expectancy per year, the whiskers represent the maximum and minimum life expectancy for a specific year, the yellow line indicates the median global life expectancy.

all global region, likewise prevalence of undernourishment saw a gradual decrease, while injuries remain a bit stagnant depicting a better model of handling injuries rather contributing to a higher life expectancy as seen in Figure 17.

- To verify this result, the correlogram and correlation matrix were analyzed. According to the correlogram, Prevalence of undernourishment has a +0.7 correlation with life expectancy, while sanitation and life expectancy had a -0.69. What drove the increase in life expectancy had to do with tackling undernourishment and sanitation. As seen in Figure 18.
- Global life Expectancy has trended upwards from 2001 to 2019 with an increase of roughly around 6 years as seen in Figure 20.
- Global life Expectancy in the year 2016 showed the lowest average to be 61.5 and the highest average of 80.6 as seen in Figure 21.

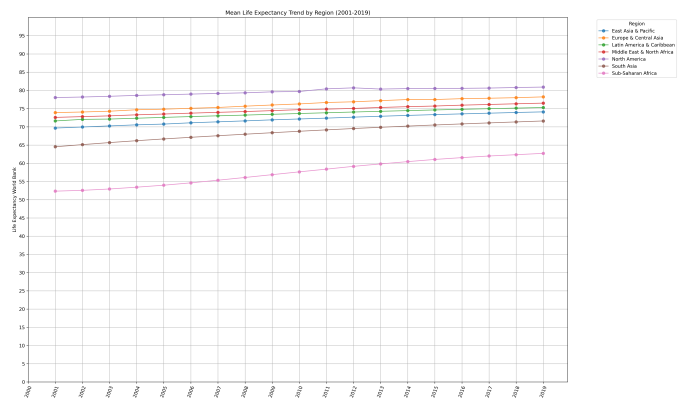


Fig. 17. Multiple line graph of global life expectancy per region showing an upward trend in global life expectancy from 2001 to 2019

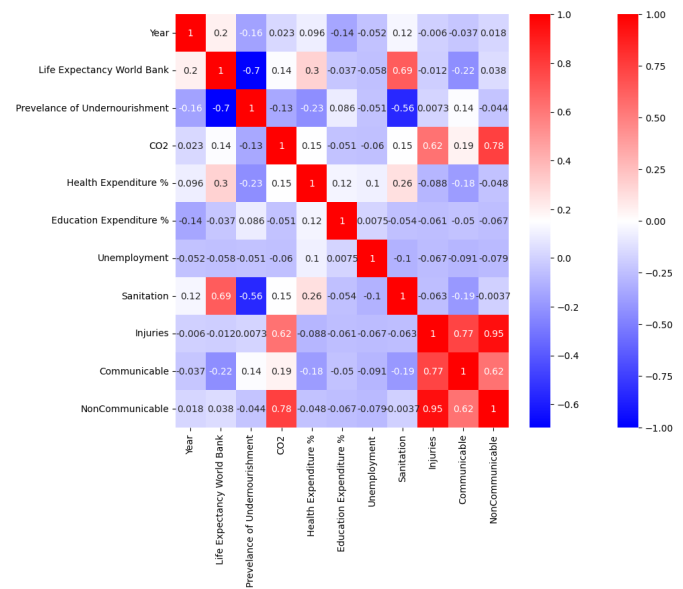


Fig. 18. Correlation matrix and score of all columns contributing to global life expectancy.

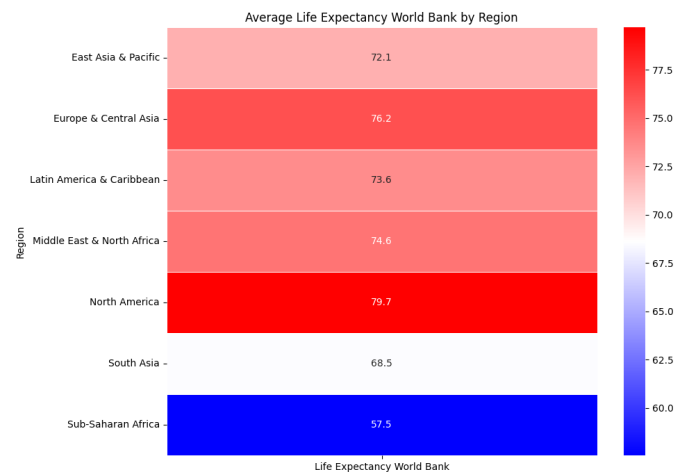


Fig. 19. Global life expectancy per region from 2001 to 2019, North America is shown to have the highest life expectancy with a median age of 79.7.

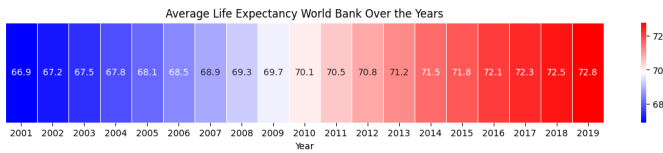


Fig. 20. Global life expectancy across all region from 2001 to 2019

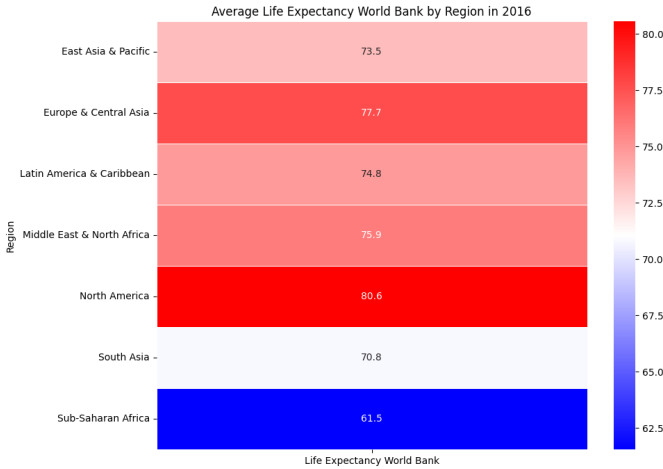


Fig. 21. Global life expectancy per region for the year 2016, North America is shown to have the highest life expectancy with a median age of 80.6.

- Finally, a chart showing the distribution of global income status across the world. From the pie chart, Europe and Central Asia has the highest amount of high income, while sub Saharan Africa has the highest amount of low income as seen in Figure 22.

B. Predictive Modeling Results

Training and Testing Dataset

Test split done to train and predict longevity was based on the standard 80:20 train split. Thereby 3306 rows were trained and derived a result of 662.

Histogram shows that the best test split is not the standard 80:20 but between 75:25 as seen in Figure 23.

Training and Validation Loss

KERAS is a poor model for predicting life expectancy due to one hot encoding categorical variable. Due to high cardinality ratio of some categorical variables such as country or country code which involves splitting each text to a binary product, this can affect and diffuse the training leading to a low R-square score (0.239) making this prediction not reliable. An alternative improvement is to drop the categorical variable and use just the numerical variable for regression before mapping the country back to the data-frame, this ensure the result is more accurate and precise.

Alternately as seen in Table VIII, the actual and predicted life expectancy sampled from various regions and different year have a low error score (MAE) and percent error signifying the prediction using random forest tree based model with a

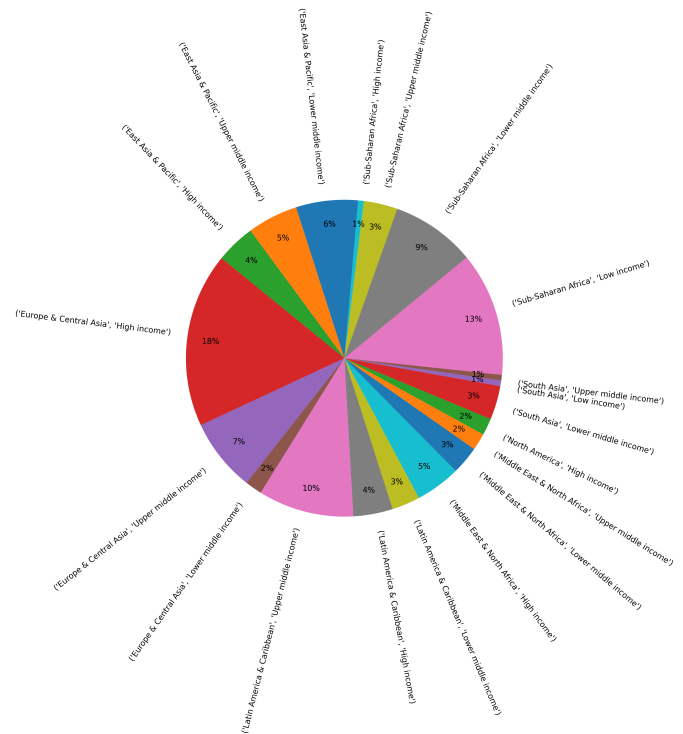


Fig. 22. Pie Chart depicting the percent of income status globally.

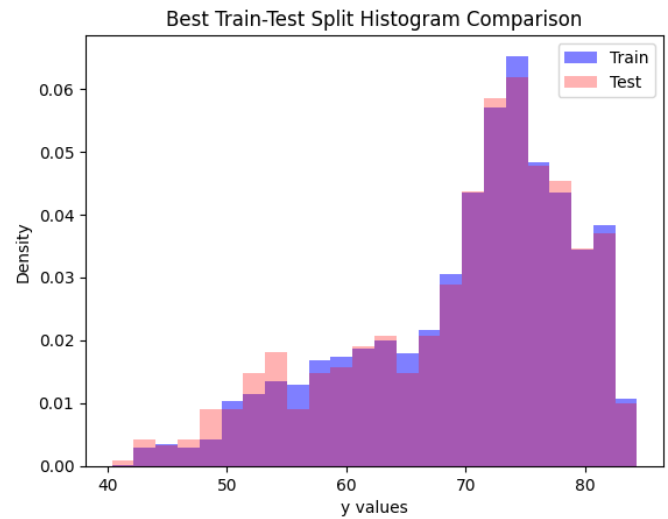


Fig. 23. Histogram depicting the preferred training to testing data split ration

max depth of 10 does a good job in predicting global life expectancy.

SHAP: Feature dominance Sub-Saharan Africa region was shown to feature more in making prediction as seen from the SHAP results as seen in Figure 25.

SHAP waterfall is used for explaining individual predictions. It starts from the model's base value (69.054) and shows how each feature's SHAP value pushes the prediction up (red) or down (blue) to reach the final output.

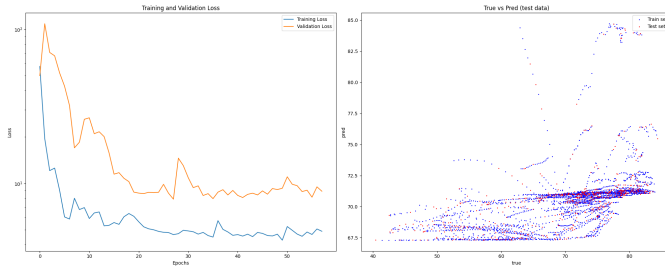


Fig. 24. To the left: Training and Validation loss curve used to determine overfitting or underfitting. To the right: True vs Pred data used to predict longevity

TABLE I
FEATURE SCORE IN PREDICTING LONGEVITY USING RANDOM FOREST REGRESSION

Features	Scores
Region_Sub-Saharan Africa	0.617941
Prevelance of Undernourishment	0.124489
Communicable	0.059312
Year	0.041327
Health Expenditure %	0.02671
Sanitation	0.020962
IncomeGroup_Lower middle income	0.013522
NonCommunicable	0.012608
O2	0.012282
Unemployment	0.011821
IncomeGroup_Upper middle income	0.010102
Injuries	0.005623
Education Expenditure %	0.003409
Region_Latin America & Caribbean	0.003046
Region_Middle East & North Africa	0.002884
Country Name_Turkmenistan	0.001282
Country Code_TKM	0.00127
Country Name_Eswatini	0.001121
Country Code_SWZ	0.001021
Country Name_Costa Rica	0.001001

Red bars (positive SHAP values) indicate features that pushed the prediction higher than the base value. Blue bars (negative SHAP values) indicates features that pushed the prediction lower than the base value. Length of the Bars: The length of each bar shows the magnitude of the feature's impact on the prediction.

Based on the SHAP value the region Sub Saharan Africa is lowering the prediction.

SHAP waterfall visualizes the importance of features in predicting longevity by attributing a SHAP value.

SHAP decision plot indicates how an individual prediction is reached, much by adding the sum of SHAP values from a baseline to the final prediction

The feature Sub-Saharan Africa is shown to contribute the most to the prediction of longevity with a score of 0.6179. Prevalence of undernourishment was second with a score of 0.1244, result derived from TABLE III.

VIF is used to check for multicollinearity among each feature to determine if they are independent of one another. A VIF score of 1 represents perfect multicollinearity meaning each features are independent of each other and do not affect their own outcome. A VIF of less than 5 is often ideal. Based on

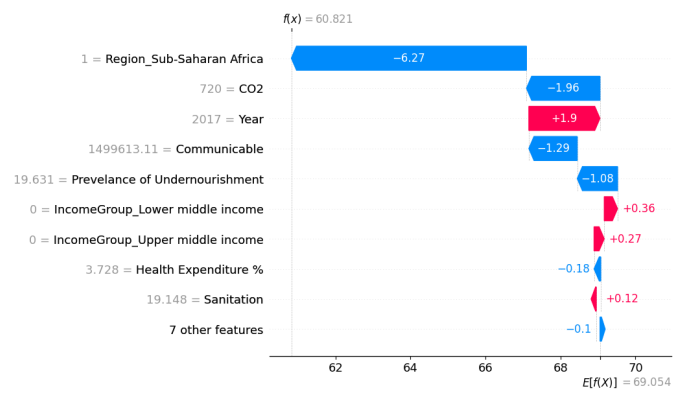


Fig. 25. SHAP Summary Plot

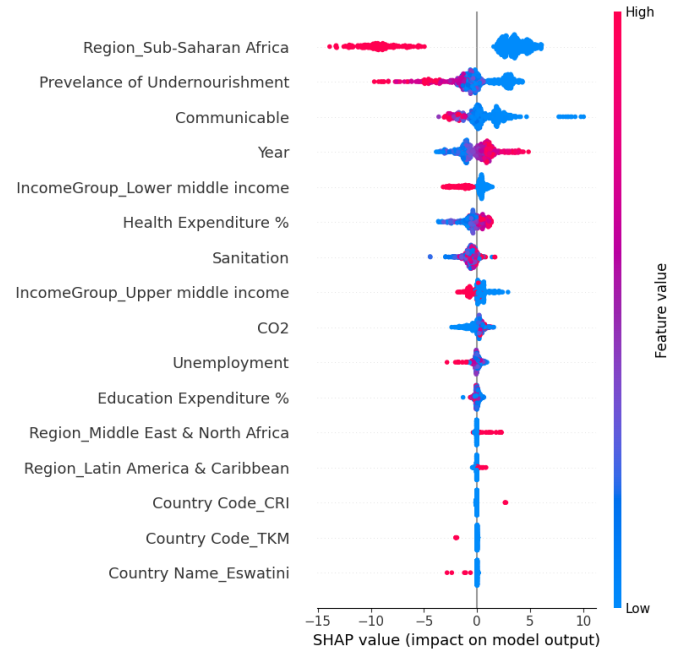


Fig. 26. SHAP Waterfall Plot

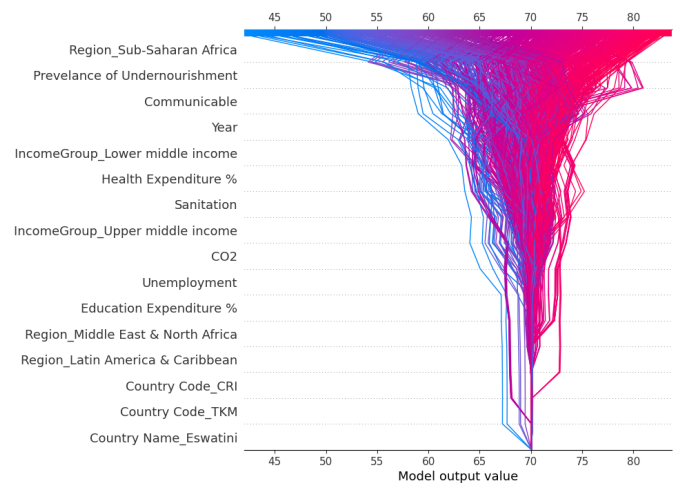


Fig. 27. SHAP Decision Plot

TABLE II

GRIDSEARCH CV FOR OPTIMIZING THE HYPERPARAMETER FOR BEST TRAINING AND TESTING THE DATASET AFTER DROPPING VIF THAT IS GREATER THAN 6

Model	Best Score	Best Parameter
KNeighbors Regressor	0.0452246418	n_neighbors': 5,
		'weights': 'distance'
Random Forest Regressor	-0.016115088	'max_depth': 10,
		'min_samples_leaf': 1,
		'min_samples_split': 2,
		'n_estimators': 50
Gradient Boost Regressor	-0.011106925	'learning_rate': 0.2,
		'max_depth': 5,
		'min_samples_leaf': 5,
		'n_estimators': 200
Decision Tree Regressor	-0.0194729265	'max_depth': 10,
		'min_samples_leaf': 1,
		'min_samples_split': 2
Linear Regression	-0.0480517862	'fit_intercept': True
XGBoost	-0.01225497	'learning_rate': 0.2,
		'max_depth': 5,
		'min_child_weight': 1,
		'n_estimators': 100

TABLE III

GRIDSEARCH CV FOR OPTIMIZING THE HYPERPARAMETER FOR BEST TRAINING AND TESTING THE DATASET BEFORE DROPPING VIF

Model	Best Score	Best Parameter
KNeighbors Regressor	-0.008555456	n_neighbors': 1,
		'weights': 'uniform'
Random Forest Regressor	-0.01560265	'max_depth': 10,
		'min_samples_leaf': 1,
		'min_samples_split': 2,
		'n_estimators': 100
Gradient Boost Regressor	-0.008479312	'learning_rate': 0.2,
		'max_depth': 5,
		'min_samples_leaf': 1,
		'n_estimators': 200
Decision Tree Regressor	-0.020342094	'max_depth': 10,
		'min_samples_leaf': 1,
		'min_samples_split': 2
Linear Regression	-0.013728304	'fit_intercept': True
XGBoost	-0.01230433	'learning_rate': 0.2,
		'max_depth': 5,
		'min_child_weight': 2,
		'n_estimators': 100

TABLE IV

VIF SCORE TO CHECK FOR MULTICOLINEARITY AMONG VARIABLES AFTER DROPPING VIF THAT IS GREATER THAN 6

Features	VIF
Region_Sub-Saharan Africa	3.111403
Prevelence of Undernourishment	1.84216
Communicable	1.195477
Year	1.093924
Health Expenditure %	1.309959
Sanitation	3.14268
IncomeGroup_Lower middle income	1.565285
CO2	1.132919
Unemployment	1.199258
IncomeGroup_Upper middle income	1.856175
Education Expenditure %	1.112672
Region_Latin America & Caribbean	1.448923
Region_Middle East & North Africa	1.18235
Country Code_TKM	1.040373
Country Code_SWZ	1.102824
Country Name_Costa Rica	1.040506

TABLE V

MANUAL FOLD FOR KERAS K-FOLD VALIDATION BASED ON OPTIMIZED HYPERPARAMETERS TO DETERMINE WHICH FOLD IS BEST USED FOR RUNNING THE MODEL.

Fold	MAE	R ²
Fold 1	Train: 70.0054	Train: -58.4627
	Test: 69.7780	Test: -55.8897
Fold 2	Train: 69.9353	Train: -58.1512
	Test: 70.0581	Test: -57.0517
Fold 3	Train: 69.8265	Train: -57.3985
	Test: 70.4934	Test: -60.3568
Fold 4	Train: 69.9701	Train: -57.3492
	Test: 69.9186	Test: -60.3583
Fold 5	Train: 70.0619	Train: -58.3039
	Test: 69.5514	Test: -56.5789

the feature score, the region sub-Saharan Africa was shown to have a VIF of 3.11 while prevalence of undernourishment is depicted to have a VIF score of 1.842. VIF greater than 6 was removed thereby dropping the feature noncommunicable (37.68) and injuries (35.38) as seen in TABLE IV.

Predicted Life Expectancy using Machine Learning Tools

Below is a summary of few countries and their predicted life expectancy using both KERAS and Random Forest as seen in Table VII and Table VIII. Random Forest predictions are more accurate due to an extreme low error and percent error.

TABLE VI
COMPARISON OF DEEP LEARNING KERAS WITH SCI KIT LEARN
REGRESSION MODEL TO PREDICT LIFE EXPECTANCY

Model	MAE	R ²
Linear Regression	Train: 3.1223 Test: 3.1408	Train: 0.8092 Test: 0.8100
KNeighbor Regressor	Train: 2.3140 Test: 3.1183	Train: 0.8379 Test: 0.7261
KNeighbor Regressor (n_neighbors=10)	Train: 2.9085 Test: 3.4517	Train: 0.7790 Test: 0.6910
KNeighbor Regressor (n_neighbors=50)	Train: 4.1705 Test: 4.2948	Train: 0.6372 Test: 0.5988
RandomForestRegressor (max_leaf_nodes=3)	Train: 3.9317 Test: 3.9526	*Train: 0.6967
RandomForestRegressor (max_leaf_nodes=10)	Train: 2.7217 Test: 2.8832	Train: 0.8579 Test: 0.8456
RandomForestRegressor (max_leaf_nodes=30)	Train: 1.9248 Test: 2.1128	Train: 0.9287 Test: 0.9123
GradientBoostingRegressor (max_leaf_nodes=3)	Train: 2.3955 Test: 2.5071	Train: 0.8908 Test: 0.8793
GradientBoostingRegressor (max_leaf_nodes=10)	Train: 1.5459 Test: 1.7734	Train: 0.9530 Test: 0.9393
DecisionTreeRegressor (max_leaf_nodes=3)	Train: 3.9427 Test: 3.9709	Train: 0.6947 Test: 0.6934
DecisionTreeRegressor (max_leaf_nodes=10)	Train: 2.8455 Test: 3.0134	Train: 0.8431 Test: 0.8293
XGBRegressor (max_leaf_nodes=10)	Train: 2.1416 Test: 2.2796	Train: 0.9124 Test: 0.9018

TABLE VII
PREDICTED LIFE EXPECTANCY AND ACTUAL LIFE EXPECTANCY
SHOWING THE DIFFERENCE ERROR AND PERCENTAGE ERROR FROM
KERAS DEEP LEARNING.

Country	Year	Actual	Predicted	Error	% Error
Germany	2016	80.99	81.24	-0.25	0.31
Barbados	2008	78.18	76.32	1.85	2.37
Lebanon	2003	75.96	72.06	3.90	5.14
Albania	2010	76.56	72.43	4.12	5.39
Niger	2009	56.55	54.84	1.71	3.02
Tuvalu	2011	72.38	72.18	0.19	0.27

V. CONCLUSION FUTURE WORKS

The global trend observed in life expectancy points to an increase in longevity globally. However, there are still significant room for improvement across all regions. The two main factors as seen in the correlation matrix with strong correlation with longevity are sanitation (-0.69) and undernourishment (0.7). Data shows that this two important metrics varies across region and even within the same region, low level of undernourishment despite sanitation as a constant variable affects longevity across the board. Sanitation had a weak correlation (0.26) with health care expenditure per GDP and a weak correlation with communicable diseases (0.19). However when communicable disease or non-communicable disease is

TABLE VIII
PREDICTED LIFE EXPECTANCY AND ACTUAL LIFE EXPECTANCY
SHOWING THE DIFFERENCE ERROR AND PERCENTAGE ERROR FROM
RANDOM FOREST MAX DEPTH 10 FROM GRIDSEARCH CV PARAMETER

Actual	Predicted	Country	Year	Error	% Error
79.74	79.09	Canada	2003	0.64	0.80
71.50	71.11	Indonesia	2018	0.39	0.54
69.51	71.11	Belize	2003	-1.60	2.31
62.89	62.91	Haiti	2016	-0.01	0.02
50.89	51.34	Burkina Faso	2001	-0.45	0.89
72.28	72.74	Serbia	2002	-0.45	0.63
65.53	63.91	Eritrea	2017	1.62	2.47
80.50	80.49	Iceland	2002	0.005	0.007
80.95	81.61	United Kingdom	2015	-0.65	0.81
79.33	79.74	Austria	2005	-0.41	0.52
77.00	77.16	Uruguay	2012	-0.16	0.20
76.14	73.61	Sri Lanka	2014	2.52	3.31
77.68	77.07	Barbados	2003	0.60	0.78
78.19	78.00	Puerto Rico	2009	0.18	0.23
69.37	70.22	Philippines	2007	-0.85	1.22
77.97	80.77	U.A.E	2019	-2.80	3.59
74.93	74.86	Mexico	2013	0.06	0.08
79.77	78.40	Chile	2016	1.37	1.72

encountered the fatality ratio (injuries) is often high (0.77 for communicable disease) and 0.95 for (non-communicable diseases). In essence more injuries, more death.

The North America region had the highest level of longevity globally while sub-Saharan Africa had the lowest longevity.

Recommendation is each country in a region should tailor their expenditure on what correlates to longevity. Countries in the same region differ based on how much expenditure they budget on services. If undernourishment and sanitation is of major concern, they should allocate more resources towards that area rather than other metrics such as education expenditure. Benefits and programs should be consulted such as school lunch or other resources. You cannot teach a sick/hungry child.

If a region has a high life expectancy which is representative of most region. Expenditure should focus on other areas such as communicable diseases to optimize longevity and increase the upward trajectory faster. Communicable diseases and longevity show a weak correlation of (-0.22). It is worth noting that using predictive modeling to predict longevity, the most important feature in the target variable(life expectancy) was sub-Saharan Africa, this indicates that this region is contributing significantly to the prediction of life expectancy which is as a result of greater level of prevalence of undernourishment. Global interventions and aids from specialized agencies organization should focus on improving this disparity by providing resources to improve quality of life.

Limitations in this project is the lack of complete data as seen in the dataset that can skew the result (about over 20 countries were not scrapped from the world data bank by the author of the dataset, some countries missing historically tend to have one of the highest life expectancy world wide which could affect the result of which region truly have the highest longevity) of the average global life expectancy either positively or negatively.

Future work should focus on life expectancy across ethnic

culture in the same country which can skew the overall life expectancy of a country by skewing the mean. Most countries are made up of various ethnic background based on a caste system or cultural similarities. This clusters of individuals are often not fully represented during data collection as it makes accountability very subjective since specialized agencies are not fully aware of what goes on in the border of a specific country that drives the data output.

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